

SIM1002 Calculus I
Tutorial 2
Limit of a Function and Limit Laws

Exercise 2.2

1. Explain why the limits do not exist for the following.

(a) $\lim_{x \rightarrow 0} \frac{x}{|x|}$

(b) $\lim_{x \rightarrow 1} \frac{1}{x-1}$

2. Suppose that a function $f(x)$ is defined for all real values of x except $x = c$. Can anything be said about the existence of $\lim_{x \rightarrow c} f(x)$? Give reasons for your answer.
3. Suppose that a function $f(x)$ is defined for all x in $[-1, 1]$. Can anything be said about the existence of $\lim_{x \rightarrow 0} f(x)$? Give reasons for your answer.

For No 4-9, find the limits of the following.

4. $\lim_{h \rightarrow 0} \frac{3}{\sqrt{3h+1}+1}$

5. $\lim_{v \rightarrow 2} \frac{v^3-8}{v^4-16}$

6. $\lim_{x \rightarrow 4} \frac{4x-x^2}{2-\sqrt{x}}$

7. $\lim_{x \rightarrow -2} \frac{x+2}{\sqrt{x^2+5}-3}$

8. $\lim_{x \rightarrow 0} \frac{1+x+\sin x}{3 \cos x}$

9. $\lim_{x \rightarrow -\pi} \sqrt{x+4} \cos(x+\pi)$

10. Suppose $\lim_{x \rightarrow c} f(x) = 5$ and $\lim_{x \rightarrow c} g(x) = -2$. Find

(a) $\lim_{x \rightarrow c} f(x)g(x)$

(b) $\lim_{x \rightarrow c} 2f(x)g(x)$

(c) $\lim_{x \rightarrow c} (f(x) + 3g(x))$

(d) $\lim_{x \rightarrow c} \frac{f(x)}{f(x)-g(x)}$

11. If $2 - x^2 \leq g(x) \leq 2 \cos x$ for all x , find $\lim_{x \rightarrow 0} g(x)$.

12. It can be shown that the inequalities

$$1 - \frac{x^2}{6} \leq \frac{x \sin x}{2 - 2 \cos x} \leq 1$$

hold for all values of x close to zero. What, if anything, does this tell you about

$$\lim_{x \rightarrow 0} \frac{x \sin x}{2 - 2 \cos x}?$$

Exercise 2.3

Prove the limit statements in Q13 – Q16.

13. $\lim_{x \rightarrow 4} (9 - x) = 5.$

14. $\lim_{x \rightarrow 4} \sqrt{x - 5} = 2.$

15. $\lim_{x \rightarrow 1} \frac{1}{x} = 1.$

16. $\lim_{x \rightarrow 1} f(x) = 2$ if $f(x) = \begin{cases} 4 - 2x, & x < 1 \\ 6x - 4, & x \geq 1 \end{cases}.$